

# CSCE 5300: Introduction to Big Data and Data Science

## Chapter 1 (1.1): Overview of the Field. Technical Challenges



# Qutline



- **1.1.1. Big data analytics**
- **1.1.2. Big data challenges**
- **1.1.3. Big data characteristics**
- **1.1.4. Sources of the big data deluge**
- **1.1.5. Types of big data structures**
- **1.1.6. State of the practice in big data analytics**

# 1.1.1. Big Data Analytics



- The volume of data that one has to deal has exploded to unimaginable levels in the past decade, and at the same time, the price of data storage has systematically reduced.
- **Private companies and research institutions capture terabytes of data** about their users' interactions, business, social media, and also sensors from devices such as mobile phones and automobiles.
- The challenge of this era is to make sense of this sea of data [1], [2].

- [1] Rajkumar Buyya, Rodrigo N. Calheiros, and Amir V. Dastjerdi, *Big Data. Principles and paradigms*. Elsevier, 2016.
- [2] *Data Science & Big Data Analytics. Discovering, Analyzing, Visualizing and Presenting Data*, John Wiley & Sons, 2015.

# 1.1.1. Big Data Analytics

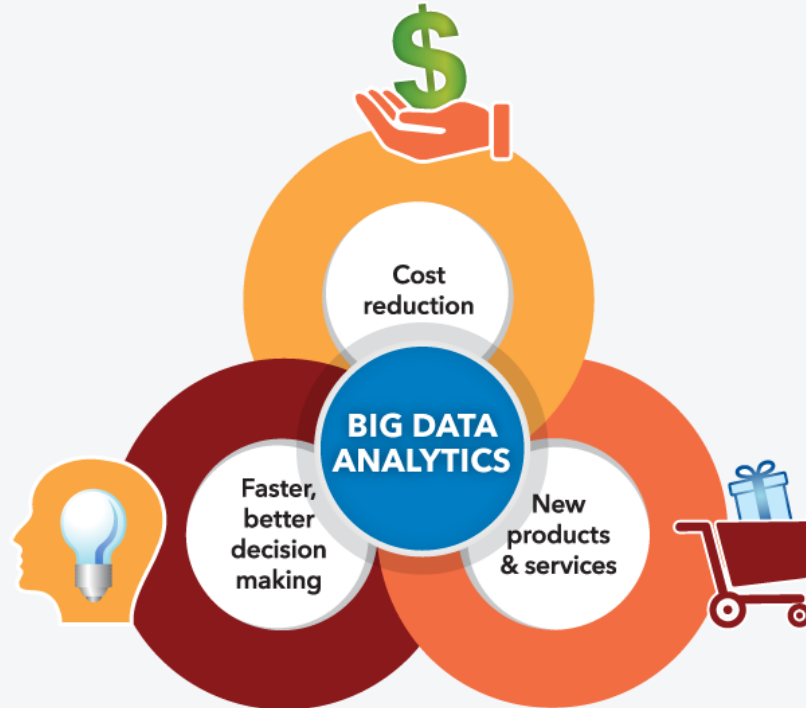


- This is where **big data analytics** is actual and used.
- *Big Data Analytics largely involves collecting data from different sources, manage it in a way that it becomes available to be consumed by analysts and finally deliver data products useful to the organization business (Fig. 1.1).*
- *The process of converting large amounts of unstructured raw data, retrieved from different sources to a data product useful for organizations forms the core of Big Data Analytics.*

# 1.1.1. Big Data Analytics



- Fig. 1.1.



## 1.1.2. Big Data Challenges



- Data is created constantly, and at an ever-increasing rate.
- ***Mobile phones, social media, imaging technologies to determine a medical diagnosis-all these and more create new data***, and that must be stored somewhere for some purpose.
- Devices and sensors automatically generate diagnostic information that needs to be stored and processed in real time.

## 1.1.2. Big Data Challenges



- *Merely keeping up with this huge influx of data is difficult, but substantially more challenging is analyzing vast amounts of it, especially when it does not conform to traditional notions of data structure, to identify meaningful patterns and extract useful information.*
- *These challenges of the data* deluge present the opportunity to transform business, government, science, and everyday life.
- *Several industries have led the way in developing their ability to gather and exploit data:*

## 1.1.2. Big Data Challenges



- • ***Credit card companies*** monitor every purchase their customers make and can identify fraudulent purchases with a high degree of accuracy using rules derived by processing billions of transactions.
- • ***Mobile phone companies*** analyze subscribers' calling patterns to determine, for example, whether a caller 's frequent contacts are on a rival network.
- If that rival network is offering an attractive promotion that might cause the subscriber to defect, the mobile phone company can proactively offer the subscriber an encouragement to remain in her contract.

# 1.1.3. Big Data Characteristics



- **For companies such as Linked In and Facebook, data itself is their primary product.** The valuations of these companies are heavily derived from the data they gather and host, which contains more and more real value as the data grows.
- Three attributes stand out as defining Big Data characteristics:
- **Huge volume of data:** Rather than thousands or millions of rows, Big Data can be billions of rows and millions of columns.

## 1.1.3. Big Data Characteristics



- ***Complexity of data types and structures:*** Big Data reflects the variety of new data sources, formats, and structures, including digital traces being left on the web and other digital repositories for subsequent analysis.
- ***Speed of new data creation and growth:*** Big Data can describe high velocity data, with rapid data using and near real time analysis.

## 1.1.3. Big Data Characteristics

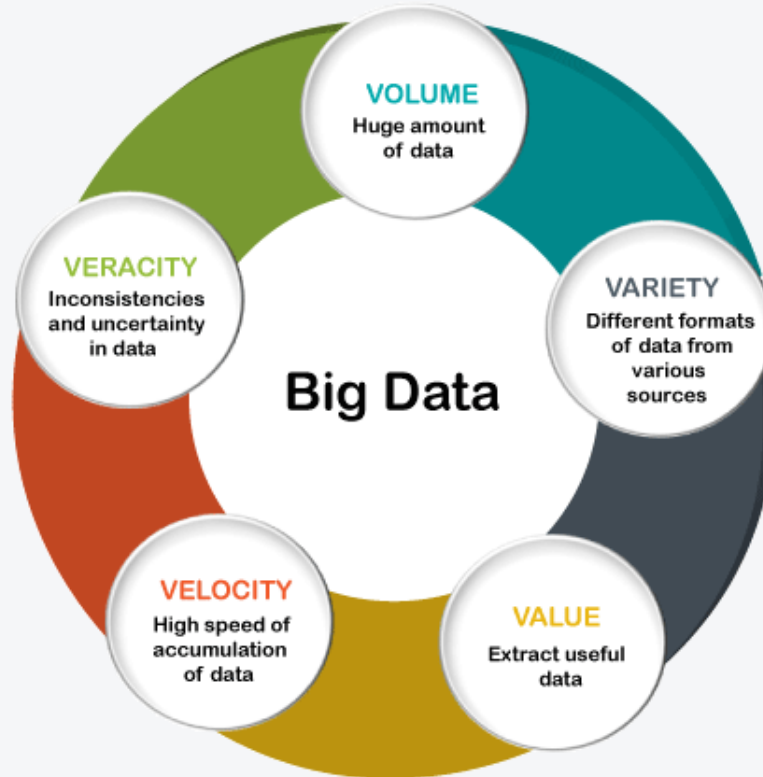


- Although the volume of Big Data tends to attract the most attention, generally the variety and velocity of the data provide a more appropriate definition of Big Data.
- Big Data is sometimes described as having volume, variety, and velocity (Fig. 1.2).
- Due to its size or structure, Big Data cannot be efficiently analyzed using only traditional databases or methods.
- ***Big Data problems require new tools and technologies to store, manage, and realize the business benefit.***

# 1.1.3. Big Data Characteristics



- Fig. 1.2.



# 1.1.3. Big Data Characteristics



- These new tools and technologies enable creation, manipulation, and management of large datasets and the storage environments that house them.
- Big Data can be defined as:
- ***Big Data is data whose scale, distribution, diversity, and/or timeliness require the use of new technical architectures and analytics to enable insights that unlock new sources of business value.***
- **Definition of Big Data implies** that organizations will need new data architectures, new tools, new analytical methods, and an integration of multiple skills into the new role of the data scientist.

# 1.1.4. Sources of the Big Data Deluge



- *Several sources of the Big Data deluge are (Fig. 1.3):*
- *Social Media*
- *Video Monitoring*
- *Mobile Sensors*
- *Video Payment*
- *Smart Grids*
- *Geophysical Exploration*
- *Medical Imaging*
- *Gene Sequencing*

# 1.1.4. Sources of the Big Data Deluge



• Fig. 1.3

## 1.1.4. Sources of the Big Data Deluge



- The rate of data creation is accelerating, driven by many of these items.
- ***Social media and genetic sequencing are among the fastest-growing sources of Big Data and examples of untraditional sources of data being used for analysis.***
- **For example**, in 2012 Facebook users posted 700 status updates per second worldwide, which can be leveraged to deduce latent interests or political views of users and show relevant ads.

## 1.1.4. Sources of the Big Data Deluge



- For instance, an update in which a woman changes her relationship status from "single" to "engaged" would trigger ads on bridal dresses, wedding planning, or name-changing services.
- **Another example** comes from genomics.
- Genetic sequencing and human genome mapping provide a detailed understanding of genetic image and origin.
- **The health care industry** is looking toward these advances to help predict which illnesses a person is likely to get in his lifetime and take steps to avoid them or reduce their impact through the use of personalized medicine and treatment.

## 1.1.4. Sources of the Big Data Deluge



- Such tests also highlight typical responses to different medications and pharmaceutical drugs, heightening risk awareness of specific drug treatments.
- While data has grown, the cost to perform this work has fallen dramatically.
- The cost to sequence one human genome has fallen from \$100 million in 2001 to \$10,000 in 2011, and the cost continues to drop.
- Now, websites such as 23andme offer genotyping for less than \$100.

## 1.1.4. Sources of the Big Data Deluge



- Although genotyping analyzes only a fraction of a genome and does not provide as much granularity as genetic sequencing, it does point to the fact that data and complex analysis is becoming more prevalent and less expensive to deploy.
- As described by the examples of social media and genetic sequencing, individuals and organizations both derive benefits from analysis of ever-larger and more complex data sets that require increasingly powerful analytical capabilities.
- ***Big data can come in multiple forms, including structured and non-structured data such as financial data, text files, multimedia files, and genetic mappings.***

## 1.1.4. Sources of the Big Data Deluge



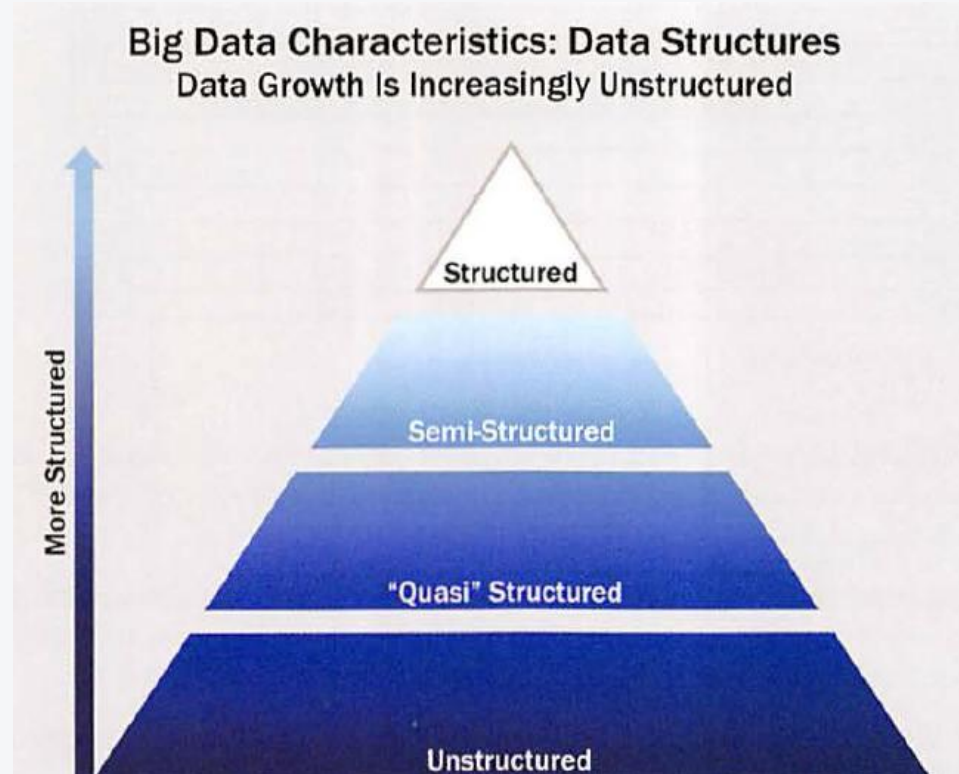
- Contrary to much of the traditional data analysis performed by organizations, ***most of the Big Data is unstructured or semi-structured in nature, which requires different techniques and tools to process and analyze.***
- ***Distributed computing environments and massively parallel processing architectures that enable using parallelized data and analysis are the preferred approach to process such complex data.***

## 1.1.5. Types of Big Data Structures



- **Four types of big data structures**, with 80-90% of future data growth coming from unstructured data types are:
  - *Structured*
  - *Semi Structured*
  - *“Quasi” Structured*
  - *Unstructured*
- Though different, the four are commonly mixed (Fig. 1.4).

# 1.1.5. Types of Big Data Structures



- Fig. 1.4

## 1.1.5. Types of Big Data Structures



- *For example, a classic Relational Database Management System (RDBMS) may store call logs for a software support call center.*
- The RDBMS may store characteristics of the support calls as typical structured data, with attributes such as time stamps, machine type, problem type, and operating system.
- In addition, the system will likely have unstructured, quasi- or semi-structured data, such as free-form call log information taken from an e-mail ticket of the problem, customer chat history, or transcript of a phone call describing the technical problem and the solution or audio file of the phone call conversation.

## 1.1.5. Types of Big Data Structures



- Many insights could be extracted from the unstructured, quasi- or semi-structured data in the call center data.
- Although analyzing structured data tends to be the most familiar technique, a different technique is required to meet the challenges to analyze semi-structured data (shown as XML), quasi-structured (shown as a clickstream), and unstructured data.
- ***Here are examples of how each of the four main types of data structures may look.***

## 1.1.5. Types of Big Data Structures



- **o Structured data:** Data containing a defined data type, format, and structure (that is, transaction data, online analytical processing [OLAP] data cubes, traditional RDBMS, CSV files, and even simple spreadsheets, (Fig. 1.5)).

## 1.1.5. Types of Big Data Structures



|    | A             | B           | C            | D          | E              | F                  | G                    |
|----|---------------|-------------|--------------|------------|----------------|--------------------|----------------------|
| 1  | Purchase ID ▼ | Last name ▼ | First name ▼ | Birthday ▼ | Country ▼      | Date of purchase ▼ | Amount of purchase ▼ |
| 2  | 1             | Davidson    | Michael      | 04/03/1986 | United States  | 10/12/2016         | 37                   |
| 3  | 2             | Vito        | Jim          | 09/01/1994 | United Kingdom | 02/02/2016         | 85                   |
| 4  | 3             | Johnson     | Tom          | 23/08/1972 | France         | 02/11/2016         | 83                   |
| 5  | 4             | Lewis       | Peter        | 18/10/1979 | Germany        | 22/11/2016         | 27                   |
| 6  | 5             | Koenig      | Edward       | 13/05/1983 | Argentina      | 26/03/2015         | 43                   |
| 7  | 6             | Preston     | Jack         | 16/06/1991 | United States  | 06/11/2016         | 77                   |
| 8  | 7             | Smith       | David        | 11/03/1965 | Canada         | 15/11/2016         | 23                   |
| 9  | 8             | Brown       | Luis         | 03/09/1997 | Australia      | 03/07/2015         | 74                   |
| 10 | 9             | Miller      | Thomas       | 07/01/1980 | Germany        | 07/11/2016         | 13                   |
| 11 | 10            | Williams    | Bill         | 26/07/1960 | United States  | 20/11/2015         | 80                   |
| 12 | 11            | Gemini      | Alexia       | 12/09/1995 | Canada         | 11/03/2017         | 35                   |
| 13 | 12            | Bond        | James        | 25/02/1975 | United Kingdom | 12/08/2017         | 40                   |
| 14 | 13            | Burgle      | Patricia     | 01/12/1990 | United States  | 18/01/2015         | 55                   |
| 15 | 14            | Reding      | Michelle     | 07/04/1985 | Canada         | 23/02/2017         | 28                   |
| 16 | 15            | Harvey      | Billy        | 14/07/1971 | United Kingdom | 12/01/2016         | 41                   |
| 17 |               |             |              |            |                |                    |                      |

Fig. 1.5.

## 1.1.5. Types of Big Data Structures



- **Semi-structured data:** Textual data files with a visible pattern that enables parsing (such as Extensible Markup Language [XML] data files that are self-describing and defined by an XMLschema (Fig. 1.6)).

# 1.1.5. Types of Big Data Structures

A screenshot of a text editor window showing an XML file named 'D:\OpenXMLTesting.xml'. The XML content is as follows:

```
<?xml version="1.0"?>
<ROOT>
  <Customers>
    <Customer CustomerName="Arshad Ali" CustomerID="C001">
      <Orders>
        <Order OrderDate="2012-07-04T00:00:00" OrderID="10248">
          <OrderDetail Quantity="5" ProductID="10"/>
          <OrderDetail Quantity="12" ProductID="11"/>
          <OrderDetail Quantity="10" ProductID="42"/>
        </Order>
      </Orders>
      <Address> Address line 1, 2, 3</Address>
    </Customer>
    <Customer CustomerName="Paul Henriot" CustomerID="C002">
      <Orders>
        <Order OrderDate="2011-07-04T00:00:00" OrderID="10245">
          <OrderDetail Quantity="12" ProductID="11"/>
          <OrderDetail Quantity="10" ProductID="42"/>
        </Order>
      </Orders>
      <Address> Address line 5, 6, 7</Address>
    </Customer>
    <Customer CustomerName="Carlos Gonzlez" CustomerID="C003">
      <Orders>
        <Order OrderDate="2012-08-16T00:00:00" OrderID="10283">
          <OrderDetail Quantity="3" ProductID="72"/>
        </Order>
      </Orders>
      <Address> Address line 1, 4, 5</Address>
    </Customer>
  </Customers>
</ROOT>
```

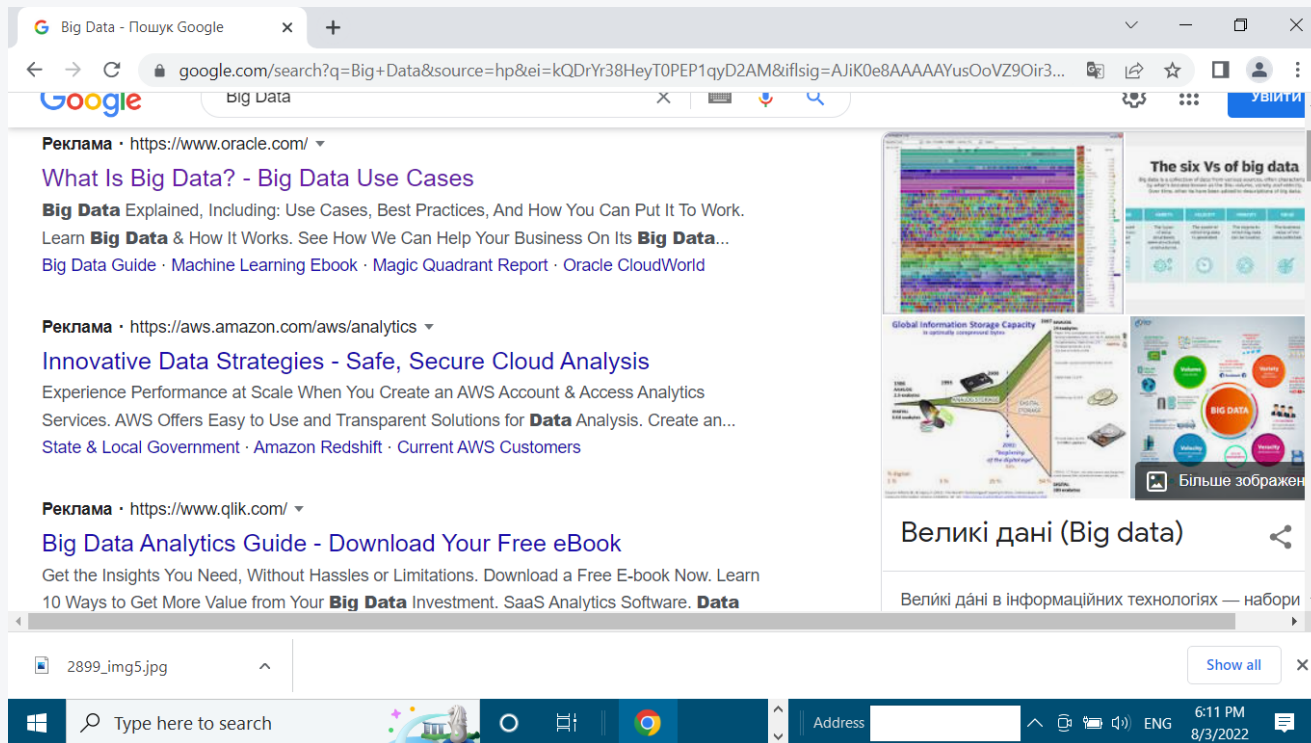
• Fig. 1.6.

## 1.1.5. Types of Big Data Structures



- **o Quasi-structured data:** Textual data with different data formats that can be formatted with effort, tools, and time (for instance, web clickstream data that may contain inconsistencies in data values and formats) (Fig. 1.7).

# 1.1.5. Types of Big Data Structures



• Fig. 1.7.

## 1.1.5. Types of Big Data Structures



- **o Unstructured data:** Data that has no inherent structure, which may include text documents, PDFs, images, and video (Fig. 1.8).

## 1.1.5. Types of Big Data Structures



- Fig. 1.8.

## 1.1.5. Types of Big Data Structures



- The four data types described above are sometimes generalized into two groups: **structured and unstructured data**.
- Big Data describes new kinds of data with which most organizations may not be used to working.

## 1.1.6. State of the Practice in Big Data Analytics



- Current business problems provide many opportunities for organizations to become more analytical and data driven, as shown in Table 1.1.

## 1.1.6. State of the Practice in Big Data Analytics



- *Table 1.1.*

| Business Driver                             | Examples                                                                |
|---------------------------------------------|-------------------------------------------------------------------------|
| Optimize business operations                | Sales, pricing, profitability, efficiency                               |
| Identify business risk                      | Customer churn, fraud, default                                          |
| Predict new business opportunities          | Upsell, cross-sell, best new customer prospects                         |
| Comply with laws or regulatory requirements | Anti-Money Laundering, Fair Lending, Basel II-III, Sarbanes-Oxley (SOX) |